

Solubility Rules for Ionic Compounds in Water

Soluble Ionic Compounds

Insoluble Exceptions

All common compounds of Group 1 ions (Li^+ , Na^+ , K^+ , etc.)

None

All common compounds of ammonium ion (NH_4^+)

None

All common nitrates (NO_3^-), acetates (CH_3COO^- or $\text{C}_2\text{H}_3\text{O}_2^-$), chlorates (ClO_3^-), and perchlorates (ClO_4^-)

None

All common chlorides (Cl^-), bromides (Br^-), and iodides (I^-)

Chlorides, bromides, and iodides of Ag^+ , Pb^{2+} , Cu^+ , and Hg_2^{2+}

All common fluorides (F^-)

PbF_2 and fluorides of Group 2

All common sulfates (SO_4^{2-})

CaSO_4 , SrSO_4 , BaSO_4 , Ag_2SO_4 , PbSO_4 , Hg_2SO_4

Insoluble Ionic Compounds

Soluble Exceptions

All common metal hydroxides

Group 1 hydroxides and $\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$, and $\text{Ba}(\text{OH})_2$

All common carbonates (CO_3^{2-}), chromates (CrO_4^{2-}), and phosphates (PO_4^{3-})

Carbonates, chromates, and phosphates of Group 1 and NH_4^+

All common sulfides

Sulfides of Group 1, Group 2, and NH_4^+
